Fully Dynamic Neural Vine Dependence and Recurrent Reinforcement Learning for Leveraged Long–Short Portfolio Allocation

Gabriel M. J. Gelissen
Fengyi Yuan
School of Science and Engineering, The Chinese University of Hong Kong, Shenzhen

Abstract

This paper develops a dependence-aware reinforcement-learning framework for monthly portfolio allocation. A training-only, data-ordered Student-t D-vine is equipped with neural correlation forecasts on all unconditional and conditional edges. Synthetic paths are calibrated to training-only empirical monthly marginals and serial dependence; realised historical data are used for fine-tuning, while the final 24 holding periods remain locked for evaluation. A recurrent TD3 agent selects self-financing long–short portfolios under common net- and gross-exposure limits. The multi-period objective is terminal-wealth CRRA represented by dense telescoping utility increments, with ex-ante CVaR, turnover, stock-borrow, and financing costs. We provide a leakage-aware, falsifiable evaluation protocol and make no superiority claim until common-return benchmarks, actual ablations, multiple training seeds, and multiple-comparison-aware inference have been completed.

Keywords: dynamic portfolio choice; vine copulas; reinforcement learning; CVaR; synthetic data; reproducible finance.

1 Introduction

Portfolio allocation is a sequential decision problem: holdings selected today affect both next-period wealth and the cost of future rebalancing. The mean–variance paradigm remains a useful benchmark, but estimation error, non-Gaussian marginal behaviour, and changing cross-asset dependence complicate its use in practice [**?** **?**]. These difficulties are particularly consequential when diversification fails in market stress.

Copulas separate marginal dynamics from dependence. A vine copula represents a multivariate copula as a collection of bivariate conditional copulas and can therefore accommodate asymmetric and tail-dependent relationships that are not implied by an elliptical dependence model [**?** **?** **?**]. Reinforcement learning (RL), in turn, provides a natural language for repeated portfolio choice because a policy maps an information set into an action and can account for delayed effects of turnover and wealth accumulation [**?** **?**].

Combining the two is appealing but not automatically persuasive. A flexible dependence model does not create return predictability; an RL policy trained on artificial paths can overfit the generator; and a single short backtest cannot establish superiority. The contribution of this paper is consequently methodological and empirical rather than rhetorical. We implement a dependence-aware, leveraged long–short allocation policy and specify the evidence required to assess it credibly.

The implemented framework has three components. First, a training-only Student-t D-vine uses neural forecasts for all 21 unconditional and conditional correlations; monthly snapshots provide a time-indexed dependence summary without rolling refits. Second, a recurrent TD3 agent receives those summaries together with portfolio and monthly market states and produces self-financing long–short weights subject to fixed net and gross exposure. Third, fully synthetic monthly paths are used for pre-training, realised in-sample historical paths are used for fine-tuning, and the final 24 monthly holding periods are excluded from both stages and reserved for evaluation.

This paper makes four deliberately bounded contributions. It (i) documents a concrete dependence-feature state representation; (ii) specifies a training protocol that prevents the held-out period from entering marginal fitting or realised fine-tuning rewards; (iii) distinguishes synthetic distributional fidelity from economic performance; and (iv) lays out a walk-forward, benchmark, ablation, and inference design suitable for testing the framework. It does *not* claim that dependence features, synthetic pre-training, or RL outperform conventional allocation methods before the prescribed experiments are run.

The remainder of the paper reviews the relevant literature, describes the implemented system, documents the data and protocol, provides a results-reporting template and validation plan, and concludes with the changes necessary for a publishable empirical claim.

2 Related Literature and Positioning

Vine copulas are widely used when marginal distributions and dependence must be modelled separately. Pair-copula constructions make high-dimensional dependence tractable while allowing family selection at individual edges [**? ?**]. In portfolio applications, this flexibility is valuable because lower-tail co-movement, asymmetry, and conditional dependence can matter for downside risk [**? ?**]. Most such studies nevertheless transform the fitted distribution into a one-period optimisation problem.

RL portfolio research instead focuses on sequential control. Deep networks can map high-dimensional states to continuous portfolio weights, and recurrent architectures can encode information sequences [**? ?**]. Deterministic policy gradients provide a practical baseline for continuous actions [**?**]. However, impressive backtests in this literature are vulnerable to data snooping, unstable hyperparameter selection, and mismatched comparison protocols. Risk sensitivity must also be defined precisely: CVaR is a coherent tail-risk measure [**? ?**], but a reward penalty is not the same as solving a constrained risk-sensitive control problem.

Our work sits at this intersection. The implementation supplies neural correlation forecasts for every unconditional and conditional edge of a full Student-t D-vine and a two-stage training design for recurrent TD3. The core empirical question remains narrow and testable: conditional on identical long–short exposure and cost constraints, do fully dynamic copula states and synthetic pre-training improve economically meaningful out-of-sample utility over strong non-RL and RL controls? The required controls are specified in **??**.

3 Methodology

3.1 Information set and marginal filtering

Let $r_{i,t}$ be the daily log return of asset i. All prices must be finite and positive and dates must be unique and increasing. The final 24 monthly holding periods are locked before fitting. On the preceding prefix, the implementation compares parsimonious sGARCH, GJR-GARCH, and eGARCH specifications with Gaussian or skewed/standard Student innovations and AR(0)/AR(1) means. A candidate is admissible only if fixed-lag Ljung–Box diagnostics for standardised and squared residuals exceed the pre-specified one-percent gate. For persistent volatility not absorbed by this grid, a causal two-component EWMA fallback is selected by Gaussian quasi-likelihood and is subject to the same diagnostic gate.

Evaluation-period pseudo-observations use fixed training parameters and the training residual empirical CDF:

$$u_{i,t} = \frac{\mathrm{findInterval}(z_{i,t}, \{z_{i,s} : s \in \mathcal{T}_{\mathrm{train}}\}) + 1/2}{|\mathcal{T}_{\mathrm{train}}| + 1}.$$

Consequently, a future residual cannot change an earlier rank or a fitted marginal parameter.

3.2 Fully dynamic neural D-vine

A D-vine order is chosen on training pseudo-observations by exhaustive search over the 7! paths, maximising the sum of adjacent absolute Kendall correlations. A Student-t backbone retains all six trees:

$$c_t(u_1, \ldots, u_d) = \prod_{\ell=1}^{d-1} \prod_{k=1}^{d-\ell} c_{k,k+\ell|k+1:k+\ell-1,t}.$$

Every one of the $d(d-1)/2 = 21$ unconditional or conditional edges has a neural forecast for its correlation. Conditional pseudo-observations in higher trees are recursively recomputed using the dynamically predicted lower-tree copulas, rather than copied from a static vine. Each edge network uses lagged conditional uniforms, filtered innovations, log volatilities, a static shrinkage anchor, and a score feature. It is trained by chronological validation likelihood and initialised at the static correlation. Degrees of freedom remain fixed per edge for identification; tail dependence still changes with correlation. The method therefore claims dynamic dependence, not dynamic family switching.

A held-out comparison showed that a tree-one truncation lost 0.1596 average log likelihood per observation relative to the full vine ($z = 5.04$). Accordingly, the otherwise parsimonious dynamic-first-tree/static-higher-tree shortcut is rejected for these data.

3.3 Monthly synthetic paths and staged training

Training-only realised monthly log returns define empirical marginal quantiles. Neural-vine uniforms are mapped through these quantiles. A stationary Gaussian AR copula, estimated and capped on training monthly returns, propagates marginal serial dependence while neural-vine innovations retain contemporaneous cross-sectional dependence. The generator must pass marginal, moving-block correlation, lower-tail co-exceedance, and temporal fidelity gates before training can begin.

All generated paths are consumed once in pre-training. Historical fine-tuning uses all causal 30-month context plus 24-month decision trajectories before the holdout, in balanced sequential passes. The first 30 months seed the recurrent state and are not reward observations. The final 24 holding periods enter neither stage.

3.4 Portfolio constraints, costs, and objective

For actor logits x_t, define long and short budgets

$$B_L = (G + N)/2, \qquad B_S = (G - N)/2,$$

where $N = 1$ is net exposure and $G = 1.5$ is the gross cap. The differentiable two-book action is

$$w_t = B_L \,\mathrm{softmax}(x_t) - B_S \,\mathrm{softmax}(-x_t).$$

Thus $\sum_i w_{i,t} = N$, short weights are allowed, and $\sum_i |w_{i,t}| \le G$. The initial allocation is equal weight, so the first decision is not charged for a fictitious trade from zero holdings.

With asset gross returns G_{t+1}, turnover $\mathrm{TO}_t = \|w_t - w_{t-1}\|_1$, annual short-borrow rate b_s, and cash-borrow rate b_c, net portfolio gross return is

$$g_{t+1}^{\mathrm{net}} = \left[1 + w_t^\top (G_{t+1} - 1)\right] \exp\left\{-\kappa \mathrm{TO}_t - \frac{b_s \sum_i (-w_{i,t})_+ + b_c (\sum_i w_{i,t} - 1)_+}{12}\right\}.$$

The same convention is applied to every benchmark.

We optimise terminal-wealth CRRA. Dense rewards are the telescoping increments

$$R_t = U_\gamma(W_{t+1}/W_0) - U_\gamma(W_t/W_0) - \lambda\widehat{\mathrm{CVaR}}_{0.95,t},$$

with discount one. Therefore the utility component sums exactly to $U_\gamma(W_T/W_0) - U_\gamma(1)$; it is not a sequence of unrelated myopic utilities.

3.5 Recurrent TD3 agent

The observation contains normalised wealth, last monthly returns, monthly EWMA volatilities, previous CVaR, current holdings, net/gross exposure, and Kendall/tail features from every vine edge. A genuine 30-month causal burn-in seeds each LSTM sequence. Daily conditional-mean coefficients are excluded because applying them to monthly returns would mix horizons.

The actor and two critics use one-layer LSTMs with input and hidden layer normalisation. Training uses recurrent TD3: clipped double-Q targets, delayed actor updates, target-policy smoothing, replay, gradient clipping, an initial random-exploration phase, and deterministic-algorithm controls. The deterministic actor entropy term is zero by default.

3.6 Common evaluation and inference

Strategies provide ex-ante weights only. The common evaluator applies the same 24 realised asset-return vectors, exposure constraints, turnover, borrow fees, and financing costs. CAGR and arithmetic excess-return Sharpe are reported separately. Paired utility differences use Newey–West standard errors with Holm correction, and a circular moving-block White Reality Check controls selection over multiple candidate models. Repeated RL seeds measure algorithm variability; they are not independent market histories.

4 Data and Experimental Design

4.1 Asset universe and calendar

The primary file contains 2,946 daily price observations for seven proxies: the S&P 500, NASDAQ, Dow Jones Industrial Average, SSE 50, a China dividend index, ChiNext, and gold. This produces 2,945 validated daily log returns. A publication appendix must still document exact tickers, provider, currency, time-zone close, price-adjustment convention, retrieval date, and the treatment of non-synchronous US and Chinese closes.

A calendar helper constructs pairs consisting of a month-end decision date and its next available month-end holding-period end. The last price observation is therefore never treated as a decision. After the 250-day minimum history, there are 114 training holding periods and exactly 24 locked evaluation periods. The first evaluation decision is 31 July 2024 and its return ends 30 August 2024; the final decision is 30 June 2026 and its shortened holding period ends 6 July 2026. This partial final period must be reported explicitly and checked in a complete-month robustness sample when more data become available.

4.2 Locked split and baseline configuration

No evaluation observation may affect marginal fitting, empirical transforms, vine order or parameters, neural early stopping, state normalization, RL training, hyperparameter choice, benchmark choice, or diagnostic thresholds.

Table 1: Locked baseline configuration

Component	Setting
Assets	7; net exposure 1.0, gross cap 1.5
Rebalancing and decision horizon	Monthly; 24 actions
Recurrent context	30 strictly prior monthly states
Locked holdout	Final 24 non-empty holding periods
Marginals	Diagnostic-gated GARCH grid; component-EWMA fallback
Vine	Training-ordered, six-tree Student-t D-vine
Dynamic dependence	Neural correlation on all 21 edges
Synthetic pre-training	1,000 paths, empirical monthly marginals, AR copula
Historical fine-tuning	61 trajectories, eight balanced passes (488 episodes)
CVaR scenarios per state	512; robustness at 1,024 and 2,048
RL optimisation	Recurrent TD3; 64 hidden units, one LSTM layer
Costs	Turnover 0.001; annual short 0.03; cash borrowing 0.02

4.3 Synthetic-data acceptance

Synthetic data are a model diagnostic, not a performance result. Diagnostics are calculated on monthly log returns. Marginal checks use standardised errors for means and 5% tail summaries rather than relative errors in gross means near one. Pairwise correlation compatibility uses a moving-block bootstrap from the historical training months. Five-percent lower-tail co-exceedance reports the number of conditioning and joint events and exact binomial intervals; an observed zero is not interpreted as asymptotic tail independence. Temporal checks compare lag-one returns and squared returns. The generator writes all diagnostics and exits non-zero if any required gate fails.

4.4 Comparison and inference design

The final evaluator accepts strategy weights, not self-reported wealth. It applies the same realised assets, net/gross constraints, turnover, short fees, and financing to every method. The minimum benchmark set is pre-declared: equal weight, constrained sample mean–variance, DCC-GARCH, static vine, rolling vine, no-vine TD3, no-pre-training TD3, and the risk-neutral ablation. Legacy benchmark artifacts are excluded.

Development uses expanding-origin pre-holdout folds and validation windows. The final 24 periods are inspected once after configurations are frozen. Report all training seeds, compute time, CAGR, arithmetic excess-return Sharpe, volatility, maximum drawdown, 5% CVaR, terminal wealth, turnover, and short notional. Paired HAC utility tests with Holm correction and a moving-block Reality Check address serial dependence and data snooping. Twenty-four months nevertheless have low power; effect sizes and intervals are primary.

5 Results and Evidence Required for Claims

5.1 Current status

There are no publishable RL results in the supplied codebase. The existing manuscript tables that report 2018–2026 benchmark performance are not generated by the current corrected RL protocol, omit the proposed model, and should not appear in a submission. Likewise, the ablation and sensitivity scripts currently manufacture “plausible” metrics through hard-coded degradation factors and random noise. Such outputs are not simulations of the implemented model and must never be reported as empirical evidence.

This section is intentionally a results-ready reporting structure. It becomes a results section only after the corrections in ?? and the evaluation programme in ?? are completed.

5.2 Required primary table

Table 2: Out-of-sample performance on pre-specified walk-forward windows

Strategy	Ann. return	Ann. vol.	Sharpe	Max drawdown	95% CVaR	Turnover
Equal weight	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]
Sample mean–variance	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]
DCC-GARCH	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]
Rolling vine	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]
LSTM-TD3, no vine	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]
Dynamic-Vine LSTM-TD3	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]	[RESULT REQUIRED: compute]

All returns must be net of the same stated transaction-cost convention. State the annualisation rule, risk-free rate, window aggregation method, number of seeds, and whether figures are pooled across windows or computed per window first.

The primary comparison is economic rather than cosmetic. A higher Sharpe ratio accompanied by implausible turnover, materially worse drawdown, or a gain confined to one window does not establish dominance. Show wealth curves and monthly weights for every window, not only the most favourable one. Since all strategies face the same realised return path within a window, report paired uncertainty: a stationary bootstrap of the return differential or a pre-specified

HAC-adjusted test of utility/return differentials is preferable to treating repeated random seeds on one historical path as independent market observations.

5.3 Ablation and sensitivity evidence

The ablation table must be generated by retraining and evaluating actual variants under the same locked protocol:
1. remove v_t while retaining all other features;
2. omit synthetic pre-training but keep the total tuning budget fixed and disclosed;
3. set $\lambda = 0$;
4. replace the LSTM encoder with a capacity-matched feed-forward encoder; and
5. compare the corrected monthly scenario generator with any alternative generator.

Each ablation requires its own seeds and walk-forward results. Do not infer component contributions from a single run or from changes in a different benchmark. Sensitivity analysis likewise requires actual retraining or, if computationally infeasible, a clearly defined robustness subset. Hyperparameters must be selected in validation windows, not because they maximise the final holdout result.

5.4 Interpretation rules

If the full model wins robustly, the discussion should attribute the result narrowly: the evidence would support the usefulness of its particular dependence-feature representation under this asset universe, frequency, constraints, and cost specification. It would not prove that vine copulas generally improve RL allocation or that synthetic data creates alpha. If performance is neutral or negative, report that finding directly; plausible explanations include weak predictability, feature redundancy, generator misspecification, short effective history, and optimisation instability. Either outcome is scientifically informative when the protocol is clean.

6 Conclusion

This paper documents a fully dynamic neural D-vine and recurrent TD3 framework for leveraged long–short monthly allocation. Its implemented contribution is an all-tree dependence state, temporally calibrated synthetic pre-training, historical fine-tuning, terminal-wealth CRRA, and common turnover, CVaR, borrow, and financing conventions. The manuscript intentionally does not convert that implementation into an unearned performance claim.

The most important next step is empirical, not stylistic: regenerate the full synthetic bundle, run all training seeds, generate actual benchmark and ablation weights, and execute the locked common-return evaluation. Only after multiple-comparison-aware inference should the abstract or conclusion describe an economic advantage.

The framework may become a strong empirical paper if it is presented as a carefully audited test of whether explicit dependence information helps sequential allocation. Its credibility will come from negative controls, leakage resistance, realistic costs, complete diagnostics, and reproducible code–not from a single favourable backtest. Richer asset universes, asset-specific borrow costs, market impact, and modern offline-RL baselines remain extensions rather than characteristics of the present implementation.

A Reproducibility Checklist and Revision Log

Table 3: Material corrections implemented before the experiment

Previous implementation	Corrected implementation
Rolling or tree-one-only vine dynamics	Persisted full six-tree D-vine with 21 neural correlation forecasts and recursive conditional inputs.
IID synthetic monthly paths	Training-only empirical monthly marginals plus a stationary AR copula and temporal fidelity gate.
Single warning-prone marginal grid	Hard diagnostic gate and causal component-EWMA fallback for persistent volatility.
Long-only simplex	Differentiable self-financing long–short two-book action with net and gross limits.
One-period CRRA	Telescoping terminal-wealth CRRA increments plus ex-ante CVaR.
Repeated blank LSTM reset history	Thirty strictly prior causal monthly burn-in states.
DDPG single critic	TD3 twin critics, delayed actor, target smoothing, and initial random exploration.
Free leverage and inconsistent costs	Common turnover, stock-borrow, and cash-financing convention.
Simulated ablation/sensitivity metrics	Artifact-only analysis that fails when actual trained runs are absent.
Unmatched benchmark paths	Weight-only evaluator applying identical realised returns and constraints.

B Submission gate

Before replacing any `RESULT REQUIRED` field, archive the configuration, all random seeds, package versions, raw-data checksum, code hashes, marginal diagnostics, vine metadata, synthetic diagnostics, checkpoints, per-period weights, common-return logs, and machine-readable results. The replication package must recreate every table and figure without accessing the locked holdout during selection.

The remaining gate is empirical: complete at least 20 TD3 seeds and the pre-declared benchmarks and ablations; report all runs; apply the common cost contract; and report paired HAC/Holm inference and the moving-block Reality Check. Data provenance, currency, market-close alignment, corporate actions, and the shortened final July 2026 holding period must be documented.